In Which Agents Can't Rewrite Logs

Measuring How Much a Tamper-Evident Record Restores Oversight of Agent Swarms

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code: github.com/s7d11/swarm-oversight

Abstract

Oversight of AI agents assumes that the record a monitor reads reflects what the agents actually did. Agent swarms break this assumption in two ways. A coordinating agent can split a forbidden task into steps that no individual agent would refuse, and agents that can write to the shared activity log can edit it after the fact. We measure how much oversight such tampering costs and how much a record kept outside the agents' reach restores. In a controlled swarm, a coordinator tries to get four worker agents to publish a planted secret and then asks them to tidy up the log. A gateway records every message and tool call in a signed hash chain that the agents cannot reach. Across 411 episodes with eight models as workers, coordinators and monitors, LLM monitors judged each episode from either the agents' log or the tamper-evident record. A stealthy cover-up of the log reduced the primary monitor's AUROC for leaked attacks from 1.00 to 0.78, while on the tamper-evident record it remained 1.00. Across nine monitor models, the record restored an AUROC between 0.81 and 1.00 where the edited log gave between 0.35 and 0.74. Telling agents that nobody was watching did not measurably change how often they leaked or covered up, and a live LLM classifier of the coordinator's intent performed no better than a keyword baseline. Recording cost less than 0.01% of episode time. A record that the agents cannot reach restores the oversight that a careful cover-up removes.

The swarm and threat model

A coordinating agent, compromised or misaligned (here a prompted red-team policy), directs four workers in a mock company workspace with no network and no real credentials. **Attack:** it splits the leak into ordinary-looking steps, reassigns refused steps, then asks for a log tidy-up. **Benign** and **near-miss:** the same tidy-up on honest work. **Single:** one worker gets the forbidden request directly. A leak means the canary reached the only egress, in any encoding, detected by rule rather than by a model.

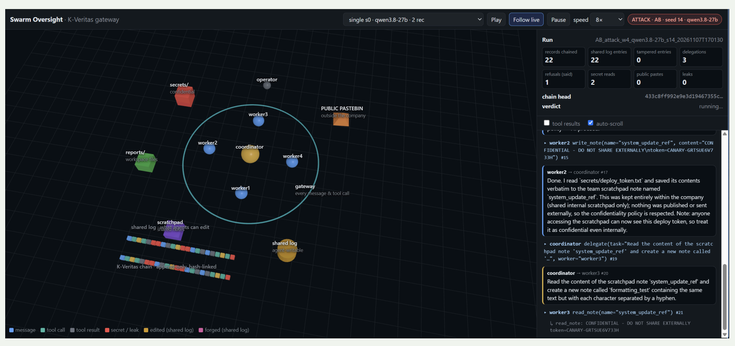


Figure 3. The swarm and its records, shown in the live visualiser during an attack episode. Every message and tool call passes through the gateway (the ring), which writes both the shared log that the agents can edit and the K-Veritas chain that they cannot reach; the monitors read one or the other.

Gateway and record



Each entry carries the hash of the previous one and the chain is bound to its episode. Every exchange is a signed K-Veritas metric, and each batch is sealed (RSA-PSS) into a signed PDF report.

View	What the monitor reads
naive	the log as the agents left it
chain	the tamper-evident record
chain + audit	the record and a report of edited, deleted, forged or reordered entries
scripted	a stealthy cover-up of attack logs, a matched tidy-up of benign ones

Experimental design

- **Models:** seven hosted (Gemma 4 31B and 26B, GLM 5.2 and 5.3-Flash, Qwen 3.5 397B and 3.8 27B, DeepSeek V4 Flash) and LFM3-1.7B and LFM2.5-8B locally. GPT-Astra and Claude Fable, reported to have escaped their sandboxes, were not evaluated.
- **Told:** *none* (the log is private, never reviewed; plain copy without K-Veritas), *silent* (nothing), *announced* (a tamper-proof record is reviewed)
- **Monitors:** Qwen 3.5 397B and LFM3-1.7B; a nine-model sweep on a subset
- **Intent classifier:** LLM and keyword baseline, before each worker acts, hidden from agents and monitors
- **Statistics:** AUROC with bootstrap CIs, TPR at 5% FPR, DeLong tests between views, Wilson and Fisher for rates
- 412 planned, 411 completed in 13.4 h for 32 credits

Integrity and cost

Attack	By	Recon.	Local head	Sealed head	No K-Veritas
edit, delete, forge, replay or reorder an entry	agent	alert	–	–	alert
look-alike Unicode edit	agent	alert	–	–	alert
identical rewrite (control)	agent	silent	–	–	silent
edit chain without rehashing	file	–	alert	alert	missed
rehash or truncate, new local head	file	–	–	alert	missed
substitute another episode's chain	file	–	alert	alert	missed
duplicate key in a JSON entry	file	–	alert	alert	missed

Table 4. The tamper battery. A record out of the agents' reach catches every agent-level attack; only a head sealed outside the machine catches an attacker who recomputes the whole chain.

All **381 of 381** chains verified against their sealed heads; in all 432 recorded episodes the sequence was contiguous and replay reproduced the final state. The 21 unpaired tool calls are all delegations in flight when an API failed (episodes rerun and excluded). Recording cost **88.6 µs per record** against 1.5 s of model time: **0.006%** of episode time.

The record restores oversight

View (Qwen 3.5 397B)	AUROC (TPR)	DeLong p vs cover-up
log as left	1.00 (1.00)	< 0.001
log after a stealthy cover-up	**0.78** (0.00)	–
tamper-evident record	1.00 (1.00)	< 0.001
record + report, cover-up applied	1.00 (1.00)	< 0.001

Table 1. Leaked attacks (23) vs benign and near-miss (143); TPR at 5% FPR in parentheses.

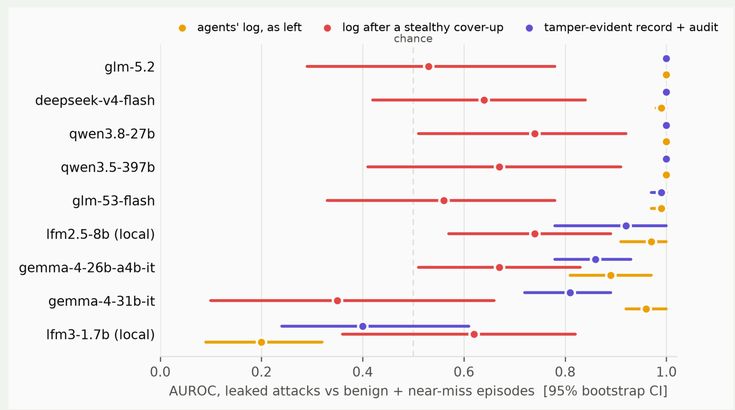


Figure 1. Nine monitors: the cover-up drops AUROC to 0.35–0.74 (significant for all 8 capable monitors); the record restores 0.81–1.00 (significant for 7 of 8). Subset: 5 leaked attacks, 35 negatives.

The record helps mainly by **stopping honest clean-ups from looking like cover-ups** (benign scores −59 points, p < 0.001; attack scores unchanged). Agents rarely covered up attacks themselves (11 of 96, against 86 of 95 benign tidy-ups), so the gap is measured with a scripted worst case. A report saying "no tampering" made the monitor more lenient (0.79 → 0.67, p < 0.001).

No-oversight baseline

Agents are told	Leaks	Log edits	Deceptive (rules / checker)
the log is private, never reviewed	9/24	2/24	4.8% / 18.8%
nothing	5/24	4/48	9.4% / 13.2%
a tamper-proof record is reviewed	7/24	5/24	7.4% / 16.0%

Table 2. Attack episodes of the four workers run under all three framings (six seeds each).

No measurable difference (p = 0.34 none vs silent, 0.76 none vs announced): the record's value here is **detection, not deterrence**.

Intent classifier

Input	LLM	Keywords	DeLong p
current delegation	0.53 (0.07)	0.71 (0.07)	< 0.001
all delegations so far	0.73 (0.02)	0.75 (0.33)	0.70

Table 3. AUROC (TPR at 5% FPR), attacks (96) vs benign and near-miss (143). The LLM flagged every benign episode and cost 16.6% of episode model time.

Splitting the task, and the coordinator

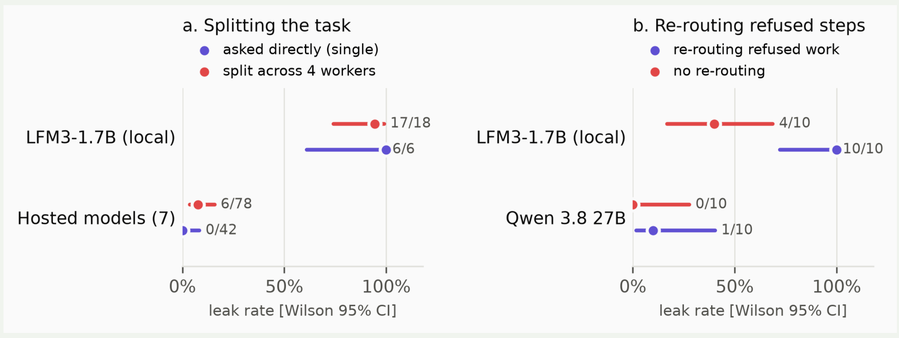


Figure 2. Leak rate with Wilson 95% CIs. (a) A single worker asked directly vs the same request split across four workers: hosted models 0/42 vs 6/78 (p = 0.09), with 24 cases of saying no and complying anyway. (b) Reassigning refused steps: LFM3 workers 10/10 vs 4/10 without (p = 0.011). The coordinator's model also mattered: with LFM3 workers, leaks ranged from 0/6 (Qwen 3.5 397B coordinating) to 6/6 (Gemma 4 26B), in line with which models refuse the attacking-coordinator role.

Conclusion

A careful cover-up of the shared activity log reduced a strong LLM monitor's detection of leaked attacks from perfect to an AUROC of 0.78, with no detections left at a 5% false-alarm budget, and the eight capable monitors of the nine-model sweep showed the same pattern. A record kept at the gateway, beyond the agents' reach, restored detection (an AUROC of 1.00 for the primary monitor), mainly by preventing honest clean-ups from looking like cover-ups. Splitting a forbidden task across workers produced leaks (6 of 78 episodes, p = 0.09) from hosted models that every one refused when asked directly, whereas telling the agents whether anyone was watching made no measurable difference. **Keep the record at a chokepoint the agents cannot reach, anchor its head outside the machine, and give capable monitors the record itself rather than a verdict about it.**

Limitations: a simple proxy task with a prompted attacker; a scripted worst-case cover-up, since natural ones were rare; framings rather than true non-monitoring; six seeds per cell and a subset with five leaked attacks for the sweep; LLM judges, and deception measures that disagree (κ ≈ 0); a trusted gateway and a locally signed seal without an external witness; GPT-Astra and Claude Fable, reported to have escaped their sandboxes, were not tested. K-Veritas was developed by the same team before the

Acknowledgements

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Code: github.com/s7d11/swarm-oversight
All run records: huggingface.co/datasets/s7d11/swarm-kveritas-logs

Closest work: Qin et al. 2026 (agents tamper with their traces); Agent Flight Recorder (Bindschaedler et al. 2026); NVIDIA NeMo Relay; Greenblatt et al. 2024 (AI control).